

Demonstration of Proposed Features

Date	10 July 2026
Team ID	
Project Name	OptiCrop : Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Crop Recommendation Prediction	Predicts the most suitable crop using soil and environmental parameters.	IMPLEMENTED	YES	Successfully demonstrated using Random Forest model.
2	Machine Learning Model Integration	Flask backend loads the trained Pickle model and generates predictions.	IMPLEMENTED	YES	Model loaded successfully during testing.
3	Input Validation	Validates all user inputs on both client and server sides.	IMPLEMENTED	YES	Prevents invalid agricultural parameter values.
4	Crop Information Display	Displays recommended crop, confidence score, water requirement, temperature range, and description.	IMPLEMENTED	YES	Information shown dynamically after prediction.
5	Responsive Web Interface	Responsive Web Interface	IMPLEMENTED	YES	Responsive across desktop browsers.
6	Crop Images	Display images of recommended crops along with results.	PENDING	NO	Planned as a future enhancement.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	5
Total Features Demonstrated	5
Overall Implementation Rate (%)	83.33%